MUHAMMAD ABDULLAH

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EDUCATION

FAST-NUCES, LHR BS Computer Science | 2023 – 2026
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Artificial Intelligence, Machine Learning, Object-Oriented Programming, Software Engineering

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript (ES6+), SQL, x86 Assembly (NASM), HTML5, CSS3
Frontend: React.js, TailwindCSS, Framer Motion, Responsive UI Design, DOM Manipulation
Backend: Node.js, Express.js, FastAPI, REST API Design & Development, SMTP Email Server integration
AI/ML: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Feature Engineering, Prompt Engineering, Model Training & Evaluation, Data Preprocessing, Hyperparameter Tuning
NLP & LLMs: Large Language Models(LLMs), Retrieval-Augmented Generation(RAG), OpenAI API Integration, Hugging Face Transformers, LangChain, LlamaIndex, Transformer Models, BERT Fine-Tuning, Semantic Search
Databases: MongoDB, MySQL, SQLite, PostgreSQL, Database Design, Normalization, Query Optimization, Indexing, Schema Design
Tools: Git, GitHub, Vercel, Render Docker (Basic Containerization), Postman API Testing, Linux CLI, Jupyter Notebook, Google Colab
Core: Machine Learning, Deep Learning, CNN Architectures, RNN/LSTM Models, Recommender Systems, System Design Basics, Software Engineering Principles, Object-Oriented Programming (OOP), Data Structures & Algorithms, Software Engineering Principles, Computer Organization & Assembly Language (COAL)

PROJECTS

ProductPulse (AI-Powered Demand & Inventory System)

- Designed and built a full-stack dashboard featuring predictive demand forecasting and real-time inventory management.
- Implemented predictive analytics for demand intelligence, automating stock replenishment suggestions.
- Synchronized database updates dynamically across components for consistent inventory tracking.
- **Tech Stack:** React.js, Node.js, Express.js, MongoDB, TailwindCSS

HealSync-AI (LLM + NLP System)

- Built AI-powered medical chatbot for symptom analysis and preliminary health guidance using OpenAI API integrated with LangChain pipeline
- Designed prompt-engineered conversational workflows to generate structured, context-aware, and reliable responses across user queries
- Implemented TensorFlow-based safety validation layer to detect and filter uncertain or unsafe outputs before delivering responses
- Developed real-time conversational system with memory-enabled interactions to improve context retention and user experience
- **Tech Stack:** Python, FastAPI, Streamlit, OpenAI API, SQLite

ScholarAssistant (AI Academic Research Assistant)

- Developed an AI assistant that performs semantic search queries across arXiv and Semantic Scholar databases.
- Built a PDF document chat feature implementing Retrieval-Augmented Generation (RAG) using OpenAI API.
- Implemented citation recommendation and citation mapping to simplify literature reviews.
- Evaluated search performance using NLP metrics (ROUGE, BLEU) to optimize citation accuracy.
- **Tech Stack:** Python, FastAPI, Streamlit, OpenAI API, SQLite

TransitFlow (Desktop Bus Ticketing System)

- Developed a Python desktop bus reservation system featuring an interactive, visual seating chart.
- Integrated Stripe payment gateway for secure ticket transactions.
- Configured APScheduler for automated reminders and SMTP server for automated email receipts.
- **Tech Stack:** Python, Tkinter, SQLite, Stripe API, SMTP

Sudoku Game (Low-Level x86 Assembly)

- Developed a fully playable Sudoku game in x86 16-bit Assembly running in DOSBox.
- Implemented direct video memory rendering and custom keyboard input handling for grid navigation.
- Configured hardware timer interrupts for timer tracking and custom PC speaker sound for audio alerts.
- **Tech Stack:** x86 Assembly, MASM, DOSBox

Klondike Solitaire Game (C++ Custom Implementation)

- Implemented a complete console-based Solitaire game using fundamental data structures without Standard Template Libraries (STL).
- Built custom implementations of stack and doubly linked list classes to handle card state transitions and undo operations.
- Managed dynamic memory allocation and clean pointer cycles to prevent runtime memory leaks.
- Tech Stack: C++, OOP, Custom Data Structures

INTERESTS

Artificial Intelligence, Large Language Models (LLMs), Web Application Development, System Architecture, Open Source Collaboration, Retro Computing & Low-Level Game Development.